



C-SLOT- COMPILER DESIGN

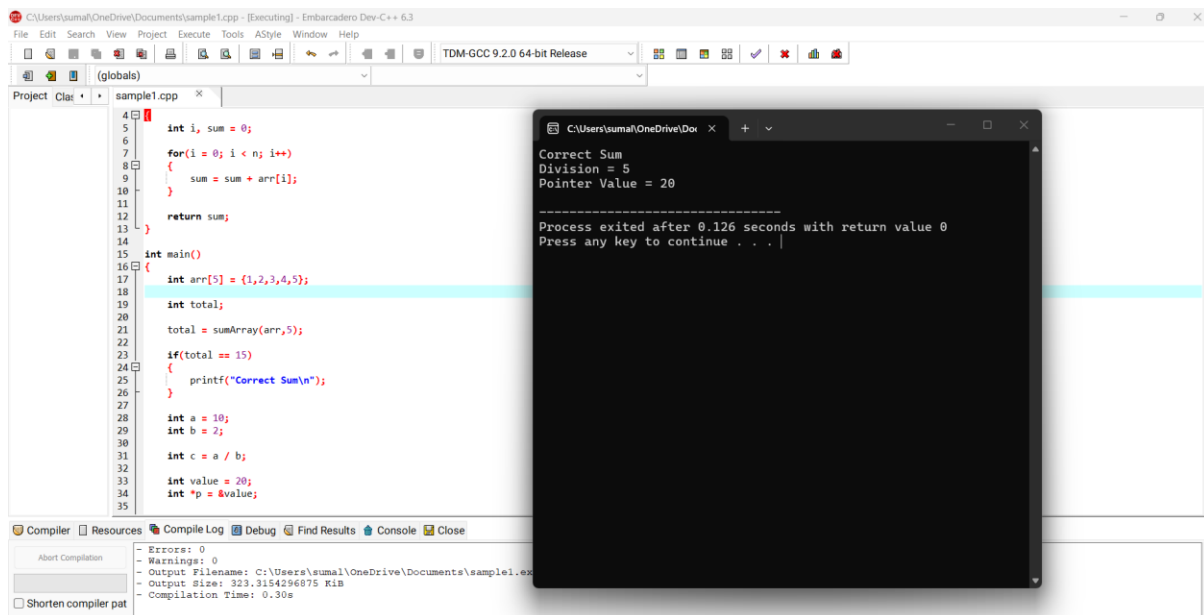
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CO1 - ASSESSMENT TOOL 2

PROGRAM 1:-

Q1. The following program is intended to perform simple arithmetic operations, function calls, array access, and looping constructs. However, the code contains multiple errors introduced at different phases of compilation, including lexical, syntax, semantic, logical, and runtime errors.

Output



The screenshot shows an IDE window titled 'sample1.cpp' with the following code:

```
1 int i, sum = 0;
2
3 for(i = 0; i < n; i++)
4 {
5     sum = sum + arr[i];
6 }
7
8 return sum;
9
10 int main()
11 {
12     int arr[5] = {1,2,3,4,5};
13     int total;
14     total = sumArray(arr,5);
15     if(total == 15)
16     {
17         printf("Correct Sum\n");
18     }
19
20     int a = 10;
21     int b = 2;
22
23     int c = a / b;
24
25     int value = 20;
26     int *p = &value;
27 }
```

The IDE also shows a console window with the following output:

```
Correct Sum
Division = 5
Pointer Value = 20

Process exited after 0.126 seconds with return value 0
Press any key to continue . . .
```

Q2. The following C program contains multiple lexical errors introduced intentionally. These errors occur due to invalid tokens, incorrect identifiers, malformed constants, and illegal symbols, which are typically detected during the lexical analysis phase of a compiler. Carefully examine the program and answer the questions given below.

Output

The screenshot shows the Embarcadero Dev-C++ 6.3 IDE. The main window displays `sample2.cpp` with the following code:

```
1 int main()
2 {
3     int num1 = 100;
4     float rate = 25.5;
5     char ch = 'a';
6     int total = 50 + 20;
7     int num_value = 30;
8     int count = 40;
9
10    printf("Welcome to Compiler Design\n");
11
12    int x = 9;
13    int y = 0x12;
14    float z = 12.34;
15
16    char str1[] = "Hello";
17    char str2[] = "World";
18
19    int _valid = 10;
20    int invalid_id = 20;
21
22    float value = 5.6;
23    int data = 10 + 5;
24
25    char c = 'x';
26    int value3 = 60;
27
28    printf("Value: %d\n", total);
29
30    return 0;
31 }
```

The console window shows the output:

```
Welcome to Compiler Design
Value: 70

-----
Process exited after 0.1221 seconds with return value 0
Press any key to continue . . .
```

The compiler status bar at the bottom indicates 0 errors and 0 warnings. The output file is `C:\Users\sumal\OneDrive\Documents\sample2.exe` with a size of 322,783,031,25 Kib and a compilation time of 0.30s.

Q3. The following C program is designed to perform arithmetic operations, function calls, conditional checks, pointer usage, and looping. However, the program contains multiple errors distributed across different phases of a compiler, including:

Output

The screenshot shows the Embarcadero Dev-C++ 6.3 IDE. The main window displays `sample3.cpp` with the following code:

```
11 float num = 15.5;
12 int result;
13
14 result = divide(10, 2);
15
16 if(result == 5)
17 {
18     printf("Result is 5\n");
19 }
20
21 int x = 10;
22 int y = 2;
23 int z = x / y;
24
25 int arr[4] = {1,2,3,4};
26 printf("%d\n", arr[3]);
27
28 int data = 25;
29 int *ptr = &data;
30 *ptr = 25;
31
32 int total;
33 total = x + y * result;
34
35 while(x < 20)
36 {
37     printf("%d\n", x);
38     x++;
39 }
40
41 return 0;
42 }
```

The console window shows the output:

```
Result is 5
4
10
11
12
13
14
15
16
17
18
19

-----
Process exited after 0.1429 seconds with return value 0
Press any key to continue . . .
```

The compiler status bar at the bottom indicates 0 errors and 0 warnings. The output file is `C:\Users\sumal\OneDrive\Documents\sample3.exe` with a size of 322,612,5 Kib and a compilation time of 0.28s.